

PROJECT: AUTOMATED BACKUP AND RECOVERY SYSTEM

Services: EC2, EBS, Lambda, CloudWatch Events. Project

Goal: Automate EBS snapshot creation and deletion using Lambda triggered by CloudWatch events. Implement snapshot retention policies.

Skills: Automation, Backup Strategies, Lambda Scripting.

Definition:

The Automated Backup and Recovery System is a cloud-based solution that automates the process of creating and managing backups for Amazon EBS volumes attached to EC2 instances. Using AWS Lambda and Amazon EventBridge (CloudWatch Events), the system automatically creates EBS snapshots on a scheduled basis and removes old snapshots according to a retention policy. This project helps ensure data protection, disaster recovery, and reduced manual effort while demonstrating skills in AWS automation, backup strategies, and serverless computing.

Scope of the project: The scope of this project is to design and implement an automated backup management system in AWS for EC2 instance storage volumes. The project focuses on scheduling EBS snapshot creation, applying retention policies to delete outdated backups, and monitoring the process using AWS services such as Lambda and EventBridge. It improves data reliability, minimizes manual administration, and provides a scalable and cost-effective backup solution suitable for cloud environments.

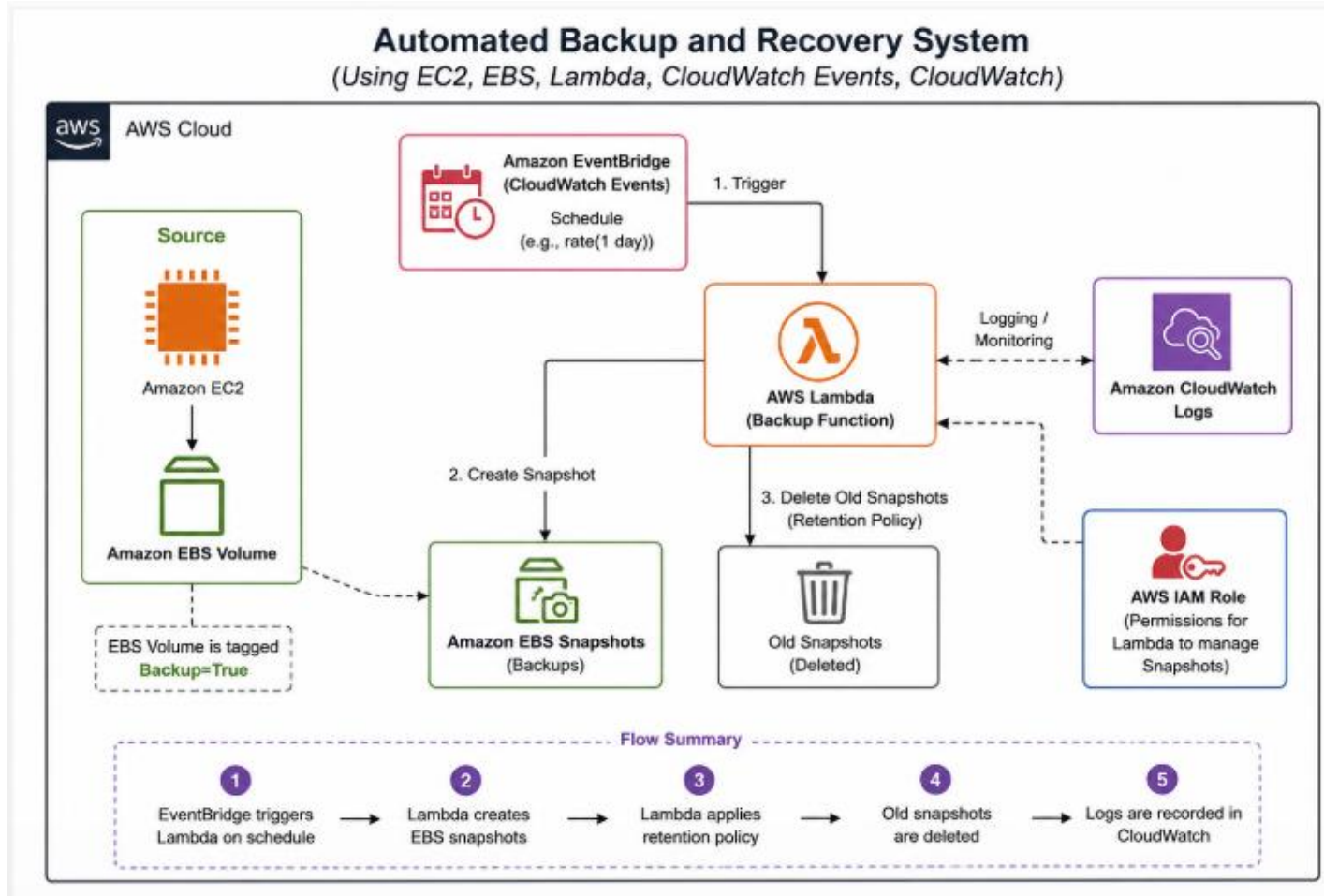
Methodologies:

1. Launch an EC2 instance with an attached EBS volume.
2. Tag the EBS volume for automated backup identification.
3. Create an IAM role with permissions for EC2 snapshots and Lambda execution.
4. Develop an AWS Lambda function to create EBS snapshots automatically.
5. Implement snapshot retention logic to delete old backups after a defined period.
6. Configure Amazon EventBridge (CloudWatch Events) to trigger the Lambda function on a schedule.
7. Monitor backup and deletion activities using CloudWatch Logs.
8. Test the system by verifying automatic snapshot creation and recovery functionality.

Services used in this project:

1. Amazon EC2 – Used to host the virtual machine and attached storage volumes.
2. Amazon EBS – Used to store data and create snapshots for backups.
3. AWS Lambda – Used to automate snapshot creation and deletion.
4. Amazon EventBridge – Used to trigger Lambda functions on a schedule.
5. Amazon CloudWatch – Used to monitor logs and backup activities.
6. AWS IAM – Used to manage permissions and secure access between services.

Architecture Diagram:



Implementation and Execution of Automated Backup and Recovery System.

Step 1: Launch EC2 Instance.

- Go to EC2 → Launch Instance
- Name: MyEC2-EBS-Project
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Project-Key)
- Click Launch Instance.

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EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types

Instances (1/1) Info Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
MyEC2-EBS-Project	i-008fada42d1de0705	Running	t3.micro	3/3 checks passed	View alarms	ap-south

Step 2: Create & Attach EBS Volume

- Created EBS volume
- Attached it to EC2

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Elastic Block Store

- Volumes**
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups

Volumes (1/2) Info Last updated less than a minute ago

Recycle Bin Actions Create volume

Saved filter sets Choose filter set Search

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source
EC2-Project-Volume	vol-0a7f4958f16377190	gp3	20 GiB	3000	125	snap-08318b8...	-

- Tag EBS Volume:
- Added tag > Key: Name > Value: EC2-Project volume

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EC2 > Volumes > vol-0a7f4958f16377190

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

Auto Scaling

- Auto Scaling Groups

Encryption Not encrypted KMS key ID - KMS key alias - KMS key ARN -

Status checks Monitoring **Tags**

Tags Manage tags

Filter tags

Key	Value
Backup	Enabled
Name	EC2-Project-Volume
Project	EBS-Automation
Owner	Shreenivasa

Step 3: Create IAM role.

- Role Name: LambdaEBSBackupRole
- Trusted Entity: Lambda
- Attached Policies: AWSLambdaBasicExecutionRole (for CloudWatch logs)

The screenshot shows the AWS IAM console page for the role 'LambdaEBSBackupRole'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, Roles, Policies, and IAM users. The main content area displays the role's summary, including its creation date (May 07, 2026, 08:57 UTC+05:30), last activity (27 minutes ago), ARN (arn:aws:iam::074139044318:role/LambdaEBSBackupRole), and maximum session duration (1 hour). Buttons for 'Delete' and 'Edit' are visible in the top right corner.

The screenshot shows the 'Permissions' tab for the role 'LambdaEBSBackupRole'. It displays a list of three attached policies: AmazonEC2FullAccess, AmazonEC2ReadOnlyAccess, and AWSLambdaBasicExecutionRole. The table below shows the details of these policies.

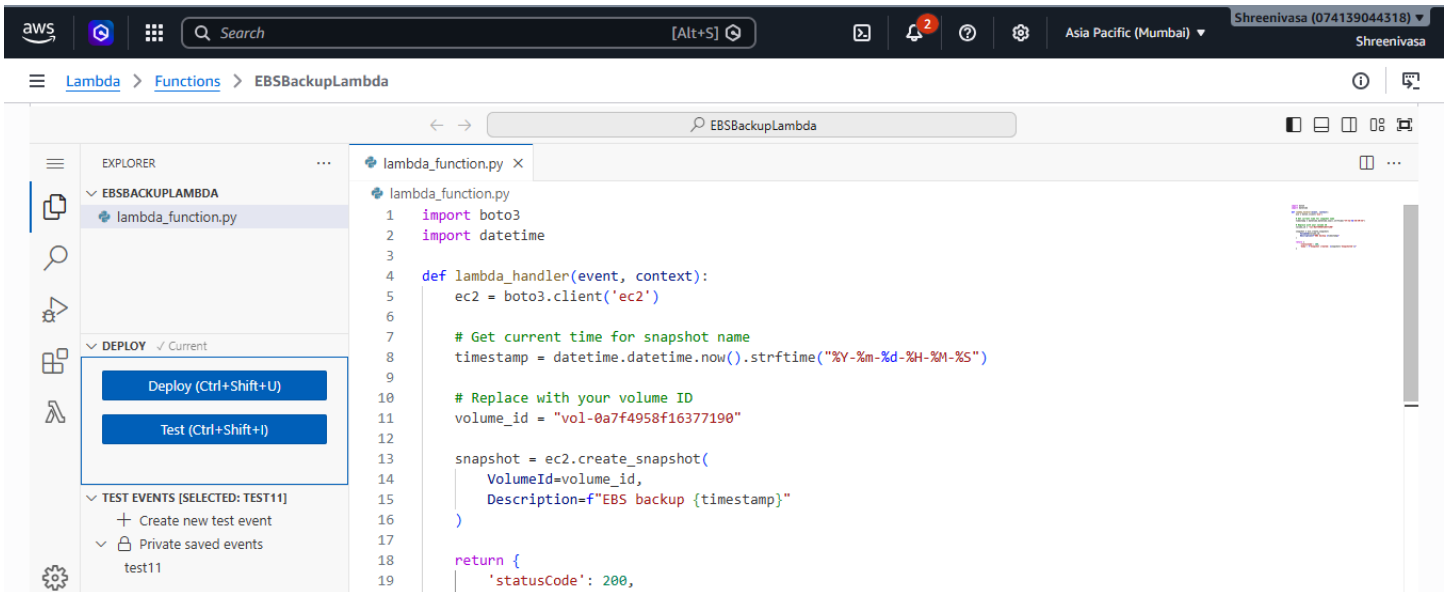
Policy name	Type	Attached entities
AmazonEC2FullAccess	AWS managed	3
AmazonEC2ReadOnlyAccess	AWS managed	1
AWSLambdaBasicExecutionRole	AWS managed	2

Step 4: Create Lambda Function

- Snapshot Creation Automation:
 - Name: EBSBackupLambda
 - Runtime: Python
 - Execution Role: LambdaEBSBackupRole

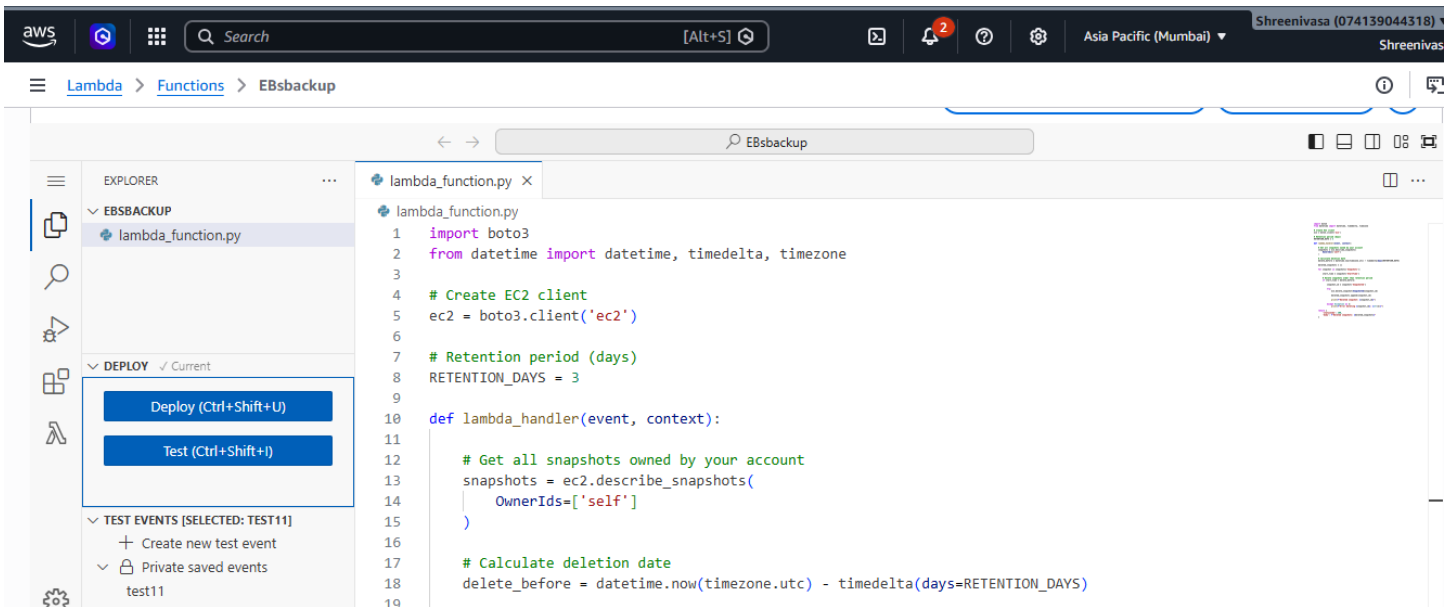
The screenshot shows the AWS Lambda console page for the 'Functions' tab. It displays a list of functions, including 'EBSBackupLambda', which was created 1 hour ago. The table below shows the details of the functions.

Function name	Description	Package type	Runtime	Type	Last modified
EBSBackupLambda	for backup project	Zip	Python 3.14	Standard	1 hour ago



• Snapshot Cleanup Automation:

- Name: EBSbackup
- Runtime: Python
- Execution Role: LambdaEBSBackupRole



Step 5: Create Event Bridge Rule – Snapshot Creation

• Created Scheduled Rule:

- Name: EBSBackupSchedule
- Schedule: rate (5 minutes) (Demo purpose)
- Target: EBSBackupLambda

Amazon EventBridge

Dashboard

▼ Developer resources

Learn

Sandbox

Quick starts

▼ Buses

Event buses

Rules

Global endpoints

Archives

Replays

▼ Pipes

You are using the new EventBridge Scheduler console. You can find your scheduled rules under Buses.

► How it works

Schedules (1)

All states

All groups

☐	Schedule name	Schedule group	Status	Target	Target type	Last modified
☐	EBSBackupSchedule	default	Enabled	EBSBackupLambda	LAMBDA_Invoke	May 07, 2026, 04:53:03 (UTC+00:00)

Step 6: Create Event Bridge Rule – Snapshot Cleanup

• Created Scheduled Rule:

- Name: EBSBackupSchedule-cleanup
- Schedule: rate (10 minutes) (Demo purpose)
- Target: EBSbackup

Amazon EventBridge

Dashboard

▼ Developer resources

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Quick starts

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Event buses

Rules

Global endpoints

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Replays

✓ Your schedule EBSBackupSchedule-cleanup is being created.

► How it works

Schedules (2)

All states

All groups

☐	Schedule name	Schedule group	Status	Target	Target type	Last modified
☐	EBSBackupSchedule-cleanup	default	Enabled	EBSbackup	LAMBDA_Invoke	May 07, 2026, 05:11:30 (UTC+00:00)

Step 7: Monitor Logs in CloudWatch

Checked:

➤ CloudWatch → Log Management → Log Groups

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CloudWatch

Log management

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log groups

Data sources - new

Summary

Get-set-go with unified data store

Log groups (2)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

1

Log group	Log class	Anomaly d...	Del...	Bea...	D...	S...
/aws/lambda/EBSBackupLambda	Standard	Configure	Off	Off	-	-
/aws/lambda/EBSbackup	Standard	Configure	Off	Off	-	-

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CloudWatch

Log management

/aws/lambda/EBSBackupLambda

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log group details

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contribut

Log streams (3)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match

Show expired

Info

1

Log stream	Last event time
2026/05/07/[\$LATEST]e107651aa3784bc99bc19575a8b3cac3	2026-05-07 10:38:40 (UTC+05:30)
2026/05/07/[\$LATEST]b6d71d104ea44186a39d696ff9020ae2	2026-05-07 10:23:47 (UTC+05:30)
2026/05/07/[\$LATEST]d41fa9d415be487785edf75cf21b2f26	2026-05-07 09:49:54 (UTC+05:30)

Snapshots are created

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Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Snapshots (9)

Info

Last updated less than a minute ago

Recycle Bin

Actions

Create snapshot

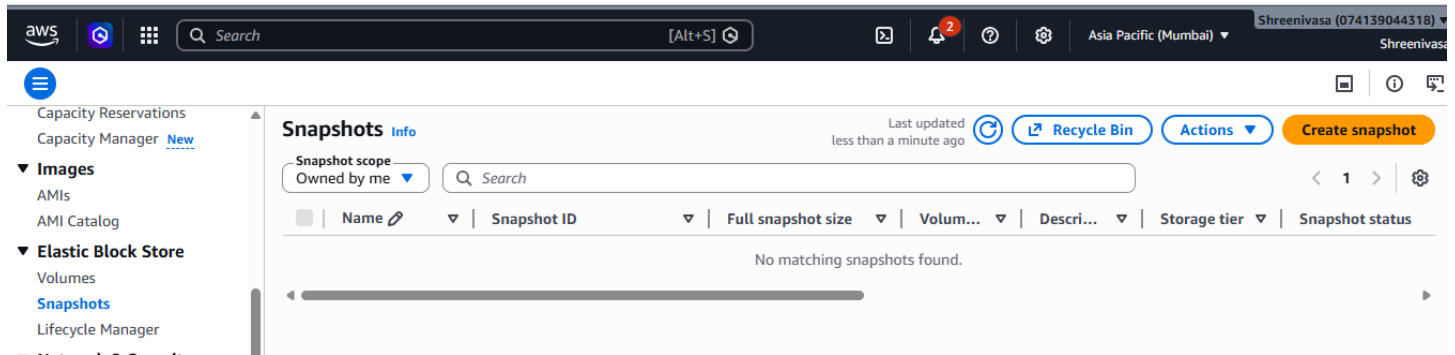
Snapshot scope Owned by me

Search

1

Name	Snapshot ID	Full snapshot size	Volum...	Description	Storage tier
	snap-016fae5b7f949ba1c	1.62 GiB	20 GiB	EBS backup 2026-05-07-04-58-43	Standard
	snap-09cbcf2afc3fe8306	1.62 GiB	20 GiB	EBS backup 2026-05-07-05-13-40	Standard
	snap-08c334a13a1262f10	1.62 GiB	20 GiB	EBS backup 2026-05-07-05-08-40	Standard
	snap-00176f66bdf1e4ed8	1.62 GiB	20 GiB	EBS backup 2026-05-07-04-53-46	Standard
	snap-0c78c4a8df1676c8c	1.62 GiB	20 GiB	EBS backup 2026-05-07-05-03-40	Standard

Snapshot is deleted:



Note:

I created one more lambda function to delete snapshots in the name of “EBSbackup” so in the code tab I edited “RETENTION_DAYS = 0” because of this snapshots are deleted immediately in one shot.

Troubleshooting:

Snapshots Accumulation During Testing

- Problem: Snapshots kept accumulating quickly due to rate (5 minutes) schedule.
- Cause: Demo schedule triggered snapshot creation frequently.
- Solution: Implemented automated cleanup Lambda and confirmed retention policy works.

Conclusion:

The Automated Backup and Recovery System successfully demonstrates how AWS cloud services can be used to automate data protection and backup management. By integrating Amazon EC2, Amazon EBS, AWS Lambda, and Amazon EventBridge, the project automatically creates and manages EBS snapshots without manual intervention. The implemented retention policy helps optimize storage usage by deleting outdated backups automatically. Overall, the project improves reliability, reduces administrative effort, enhances disaster recovery capability, and demonstrates practical skills in AWS automation and cloud infrastructure management.